Hilay Trivedi

Technical AI Lead | Senior Software Engineer | WordPress Core Contributor

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SUMMARY

Senior Software Engineer and Technical AI Lead with 5.5+ years in fully remote roles. At rtCamp (WordPress VIP Gold) I lead the internal AI initiative, reporting to the CTO, and build the systems behind it: a production knowledge retrieval layer, a persistent memory plugin for coding agents used by 80+ engineers that cut token usage per task by 32%, and a security gate for AI-written code that surfaced 129+ issues across 23 repositories. I also own discovery, scoping, and pricing for enterprise pre-sales. Outside work I build multi-agent systems with LangGraph and GraphRAG. WordPress Core contributor credited on releases 5.9 through 7.1.

SKILLS

AI / ML: LLM integration (Claude, Gemini, OpenAI), multi-agent orchestration (LangGraph, LangGraph.js), human-in-the-loop interrupts, RAG and GraphRAG, hybrid relational + vector retrieval, pgvector, Neo4j, BGE / bge-m3 embeddings, structured output, prompt-injection defense, eval design, MCP servers and clients, llama.cpp

LLM observability: Langfuse, LiteLLM, OpenTelemetry, cost and usage tracking, eval pipelines

Agentic development: Claude Code skills, hooks and plugins, CLAUDE.md / AGENTS.md instruction layers, spec-driven planning with approval gates, model-tier cost optimization, deterministic (non-LLM) metrics, Codex, Copilot

Security tooling: Semgrep custom rules, PHPStan, PHPCS (VIP), WPScan, CycloneDX SBOM, OWASP Top 10

Languages: Python, TypeScript, PHP 8+, JavaScript, SQL

Backend / infra: FastAPI, NestJS, PostgreSQL (RLS, JSONB), Neo4j, MySQL, Redis, REST, GraphQL, WebSockets, Docker, GitHub Actions, Vercel, Supabase

Frontend: React, Next.js App Router, CopilotKit, Tailwind CSS, D3.js, Three.js

WordPress VIP: Custom plugin architecture, multisite, headless (WPGraphQL + Next.js), Gutenberg / FSE, SSR blocks, ACF Pro, WooCommerce

PROFESSIONAL EXPERIENCE

rtCamp (WordPress VIP Gold Agency) — Remote

Client delivery for enterprise publishers, fintech, and government clients, plus internal AI systems.

Technical AI Lead (concurrent with Senior Software Engineer) — **Aug 2026 to Present**

- Lead rtCamp's internal AI initiative, reporting directly to the CTO; technical lead of one of two AI initiative teams. Own planning, proposals, architecture documentation, POC delivery with the team, and stakeholder presentations for org-wide adoption, with client-facing rollout planned after the POC.
- Architected the centralized company AI system: a knowledge retrieval layer as the foundation, a persistent memory layer on top, and a single queryable interface for the whole organization.
- Built and run in production the internal knowledge retrieval system over rtCamp's 16-year delivery history: ingestion and chunking pipeline, hybrid retrieval pairing a relational store (entity relationships) with a vector store (semantic similarity), hashed metadata-tagged content capsules, and answers scoped by role (sales, architecture, engineering). Self-hosted.
- Evaluated models and agent workflows against deterministic metrics, retiring 5 workflows that did not justify their cost. Authored the company AI development framework: tooling inventory, governance standards, and agent workflow guidelines.

Senior Software Engineer — **Apr 2025 to Present**

- **Persistent AI memory layer:** shipped as an internal Claude Code plugin giving coding agents shared memory, used by 80+ engineers; cut token usage per task by 32%, measured through LiteLLM per-project usage and Langfuse traces. Three-tier scoping (org / project / personal), git-backed markdown as source of truth, human-approved promotion, deterministic non-LLM reranking, per-subagent model tiers. Codex adapter in progress.
- **AI code security gate:** designed and shipped rtcamp-security-skill, checking PRs and AI-written changes against WordPress VIP standards and OWASP Top 10 across WordPress/PHP and Next.js/GraphQL repos; surfaced 129+ issues across 23 repositories. Layered enforcement (PostToolUse hooks, session-end data-flow pass, pre-commit), custom Semgrep rule pack, PHPStan, WPScan live audit, CycloneDX SBOM. Runs on demand today; CI hard-gate rollout in progress.
- **Pre-sales and growth engineering:** own discovery, solution scoping, pricing, and proposals for enterprise leads; onboarded 4 high-ticket clients in one month. A former client's Director of Engineering brought a multi-million dollar, multi-year engagement to rtCamp from his new company based on our earlier delivery together.

- Leading rtCamp's WordPress connector for the Claude Connectors Directory.
- Interview engineering candidates and mentor engineers through internal AI learning initiatives.

Web Engineer **Jan 2023 to Apr 2025**

- Led 8 engineers to ship 200+ landing pages in 6 weeks on a reusable PHP/ACF component system: 32% client sales increase, 200% daily visitor growth.
- Built the Gutenberg block system for Canada's largest media publisher: 90+ mobile / 99 desktop Lighthouse on a multisite network serving millions of readers.
- Led the revamp and redesign of a multisite network, increasing daily visits by 23%.
- Architected headless multisite networks (GraphQL + Next.js), replicating Gutenberg blocks in Next.js; built a QuickBooks integration from WordPress.

Multidots Solutions Ahmedabad

Full Stack Developer **Dec 2020 to Jan 2023**

- Led the full backend migration of a legacy .NET platform to WordPress VIP (custom plugins, WooCommerce, REST APIs), delivering a 34% post-launch sales increase.
- Built an in-house server-side rendered Gutenberg block library (PHP + React) reused across projects, accelerating team delivery.
- Modernized legacy PHP codebases and consulted directly with clients on architecture, timelines, and budget.

Llychee Inc. (Hong Kong) Remote

Machine Learning Intern **Sep 2020 to Oct 2020**

- Built ML models for an HR-tech candidate-matching system, including a LinkedIn profile scraper with ML relevance scoring against job descriptions.

SELECTED PROJECTS

Tower of Agents: governed multi-agent platform towerofagents.vercel.app

- Neo4j, GraphRAG, LangGraph, MCP, Next.js. Three layers: Neo4j content graph with GraphRAG retrieval; LangGraph governance with agent registry, HITL gates, and audit trail; MCP execution layer exposing governed capabilities as an SDK.

Advoksha: multi-agent legal research github.com/HILAYTRIVEDI/advoksha

- Next.js, FastAPI, LangGraph, Gemini 2.5, BGE, Supabase, Redis, Docker. Supervisor, Researcher, Linguistic, and Drafter agents on shared Redis state; research grounded in High Court and Supreme Court case law; OCR, regional-language translation, schema-validated contract drafting.
- Built from interviews with 15 to 20 practicing lawyers. Chose a ~32M-parameter BGE model to stay self-hostable and far below ~$250/seat incumbent pricing. Deployed, now archived.

LessonBuild: human-in-the-loop learning agent

- TypeScript, LangGraph.js, CopilotKit, Next.js, PostgreSQL, Docker. Turns a PDF into an approved lesson plan, objective-scoped MCQs, retryable feedback, and a personalized report. Postgres checkpointer makes every interrupt durable; answer-leak guardrail enforced structurally and verified with an adversarial test.

OPEN SOURCE

- **WordPress Core:** credited contributor on releases 5.9 through 7.1; 8 patches accepted in Core SVN (coding standards, date/time, site health, PHP deprecations, admin UX). Part of the initial Core AI team; proposed the Unified AI Management Layer (LLM abstraction for plugins and themes), open as a GitHub proposal.
- **Gutenberg and Meta:** color support for Query Total block, margin control for Embed Preview, accessibility and dynamic block fixes; 2 Meta SVN changesets. Plugin author: Revision Buster (WP-CLI + admin UI).

EDUCATION

Executive Diploma in Machine Learning and AI, IIIT Bangalore **Aug 2025 to Dec 2026**

Module credential earned Dec 2025. Coursework: statistics, regression, decision trees, clustering, NLP, neural networks, PySpark.

B.Tech in Computer Engineering, L.J. Institute of Engineering and Technology (CGPA 9.6 / 10) **2017 to 2021**

Diploma in Management, Ahmedabad Management Association **2022 to 2023**